

Pedro Antonio Pérez Ferrer

Computational and experimental biologist bridging wet-lab molecular biology (recombinant protein expression and purification, functional screening, BSL-2 microbiology) with computational biology (Python, long-read genome assembly, HPC pipelines, large-scale genomic data analysis). PhD in Quantitative and Systems Biology, graduating 2026.

WORK EXPERIENCE

PhD Candidate – School of Natural Sciences, University of California Merced

Jan 2021 – Current

Investigate the molecular and genomic basis of the symbiosis between *Sepiola* squids and their bioluminescent *Vibrio* symbionts as a tractable model for host-microbe interactions.

- **Pioneered** a novel *Sepiola-Vibrio* host-microbe symbiosis system, building a new computational and wet-lab analytical framework from the ground up.
- **Discovered**, described and phylogenetically classified two new *Aliivibrio* species by achieving the first high quality whole genome assembly and annotation of six different strains through long-read sequencing technology.
- **Performed** the first genetic population structure study of the *Sepiola-Vibrio* symbiosis system, integrating data from the eukaryotic host and the bacterial symbiont.
- **Solved** major biochemical bottlenecks by optimizing recombinant expression systems for difficult-to-express regulatory transcription factors (LuxR). Successfully rescued and refolded proteins from inclusion bodies, achieving high purity yields via Cytiva ÄKTA FPLC for downstream kinetic assays.
- **Managed** cross-functional international collaborations, actively coordinating genomic data analysis with a partner lab at the University College London and small molecules characterization in the Sorbonne Université in France.

References: Dr. Michele Nishiguchi, Dr. Michael Thompson

AI/ML projects (self-directed) — github.com/Sepietta

2025– 2026

Built primarily in Python/TypeScript, leveraging LLMs and Claude to solve applied biological and clinical problems.

- **symbiosis-abstract-llm**: Fine-tuned a BioBERT model on a custom-curated PubMed dataset for multi-class classification of symbiosis-related scientific abstracts.
- **food-safety-ai-pipeline**: Built an AI-powered Python pipeline automating microbiological reporting for food-safety labs (FSMA validation, FDA recall intelligence, PDF report generation).
- **clinical-trial-project**: Developed a tool for worldwide clinical-trial search (TypeScript).

Data Mentor URISE Program – University of California Merced

Jan 2026 – Current

Mentor undergraduate researchers of varying experience levels in data analysis and bioinformatics under an NIH-funded computational biology program.

- **Directed** five concurrent computational biology projects under an NIH-funded initiative, training students in modern bioinformatics workflows.
- **Mentored** interdisciplinary teams in advanced data analysis techniques, translating raw computational data into high quality outputs to be presented at a conference.
- **Provided** comprehensive technical project management and mentorship, meeting weekly with each team to ensure they meet their training requirements.

References: Dr. Michele Nishiguchi, Dr. Rudy Ortiz

Bioinformatics Research Intern - Department of Energy (DOE) Joint Genome Institute (JGI) at Lawrence Berkeley National Laboratory

May 2024 – Aug 2024

Improved the JGI's fungal mitochondrial genome annotation pipeline, resolving errors caused by circular genomes where genetic elements span the FASTA scaffold's split point.

- **Engineered** a robust troubleshooting workflow on the NERSC-HPC to resolve incorrectly annotated genomes, improving the fungal mitochondrial genome annotation pipeline of the JGI user facility.
- **Successfully** recovered and completely annotated 47% of 600 previously failed fungal genome annotations by developing automated Python and Bash scripts to identify and correct GFF3 and GFF files.
- **Scaled** the optimized *in silico* pipeline for broad deployment, integrating the new fixed pipeline in the MycoCosm database and expanding the availability of high-quality mitochondrial genomes for the scientific community.
- **Collaborated** with senior computational biologists to validate annotation accuracy and participated in the annual conference to share my results.

References: Dr. Asaf Salamov, Dr. Sajeet Haridas and Dr. Igor Grigoriev

Teaching Assistant – University of California Merced

Jan 2021- Dec 2025

Taught introductory biology and advanced microbiology across virtual and in-person formats, training students in classical and modern lab techniques and leading hands-on research projects such as the Small World Initiative for antibiotic discovery.

References: Dr. Liza Gomez-Daglio, Dr. Laura Beaster-Jones, Dr. Melinda Findlater, Dr. Petra Kranzfelder, Dr. Marcos Garcia-Ojeda

Research Intern - Sorbonne Université / Centre National de la recherche scientifique (CNRS)

Aug 2019 – Dec 2019

Investigated quorum sensing within the *Roseobacter* clade at the LBBM (CNRS / Sorbonne Université), identifying novel acyl-homoserine lactone (AHL) signaling molecules with potential applications in aquaculture disease control.

- **Processed** and annotated complex mass spectrometry datasets using MZmine to accurately annotate and calculate acyl homoserine lactone concentrations.
- **Employed** Global Natural Products Social Molecular Networking (GNPS) to map chemical diversity and identify novel quorum-sensing molecules (AHLs) with therapeutic potential for aquaculture diseases.

References: Dr. Raphaël Lami, Dr. Alice Rodrigues

Graduate Researcher – Microbial Exopolysaccharide Research Group at University of Granada Aug 2018 – Jul 2019

Identified and isolated marine bacterial strains with quorum-quenching activity, demonstrating reduced virulence-factor production in the human pathogen *Pseudomonas aeruginosa* PAO1 and the coral pathogen *Vibrio coralliilyticus*.

- **Executed** an antimicrobial discovery initiative focused on enzymes with quorum quenching activity, screening marine invertebrates' microbiomes.
- **Reduced** the virulence factor production from the highly relevant human pathogen *Pseudomonas aeruginosa* PAO1 (laboratory strain).

- **Translated** *in vitro* assay hits into *in vivo* efficacy successfully attenuating *Vibrio coralliilyticus* pathogenicity and reducing mortality in the *Artemia salina* infection model.

References: Dr. Inmaculada Llamas

Undergraduate Researcher – Microbiology Department at University of Málaga

Nov 2016 – Jul 2018

Conducted undergraduate research in bacterial characterization and identification, funded by the competitive I Research and Transference Fellowship.

- Utilized API identification protocols for newly discovered bacteria.
- Characterized lux operon using PCR.
- Established range of temperature and salinity using classical microbiology techniques.
- Performed antibiogram, resistotype and plasmidotype along with analysis of the protein profile with MALDI-TOF to fully describe the new species.

RESEARCH SKILLS

Computational Biology & Bioinformatics:

- **Workflow & Environment Management:** Nextflow, Bioconda, Git/GitHub.
- **Core Computing:** HPC, National Energy Research Scientific Computing (NERSC), MERCED Cluster, Linux/Unix (Bash/Shell), Python, R/RStudio.
- **Data Analysis & Visualization:** Python (Pandas, NumPy, SciPy, Matplotlib, Seaborn), R (Tidyverse, ggplot2, vegan, DESeq2), Statistical Modeling (PCA/PCoA, ANOVA, Hierarchical Clustering, Multivariate Statistics), GraphPad Prism.
- **Genomics & Comparative Multi-Omics:** De Novo Long-Read Assembly (PacBio HiFi), Read QC (FastQC), Rapid Genome Annotation (Bakta), Pangenomics, Biosynthetic Gene Cluster (BGC) Mining (antiSMASH), Comparative Genomics (Anvi'o), Phylogenomics.
- **Databases:** NCBI (GenBank, BLAST, SRA), PDB, AlphaFold DB, antiSMASH database / MIBiG, UniProt, Pfam/InterPro, KEGG, MycoCosm.
- **AI/ML:** LLM fine-tuning (BioBERT), prompt engineering, AI-assisted application development (Python/TypeScript); IBM AI Developer certificate; ACCESS-NAIRR AI workshops.

Wet-Lab & Analytical Chemistry:

- **Microbiology and In Vivo Models:** Aerobic & Anaerobic Cultivation, Host-Microbe Symbiosis Modeling (Vibrio-Squid), Bioluminescence Assays, Bacterial Growth Kinetics, Phenotypic Screening, BSL-2 Aseptic Technique.
- **Molecular Biology:** HMW DNA Extraction, PCR/qPCR, Plasmid Design (Snapgene, Geneious), Bacterial Transformation, Recombinant Expression Optimization (Hard-to-Express Targets), Protein and Nucleic Acid characterization.
- **Biochemistry:** Inclusion Body Solubilization & Refolding, Protein Production & Purification (Cytiva ÄKTA FPLC, UNICORN software), Chromatography (SEC, IEX, Affinity: His-trap, GST), CD, Biomolecular Interactions (Biolayer Interferometry / BLI, EMSA), SDS-PAGE, Western Blot.
- **Analytical Chemistry and Assay Development:** High-Throughput Microplate Assays (SpectraMax iD5, 96/384-well formats), HPLC-MS/MS, Molecular Networking, Small Organic Compound Discovery (Acyl Homoserine Lactones).

EDUCATION

- Ph.D., Quantitative and Systems Biology - University of California, Merced (2021-2026).
- M. Sc., Quantitative and Systems Biology - University of California, Merced (2024).

- M. Sc., Research and Advances in Microbiology - University of Granada, Spain (2019).
- Professional clinical analysis and microbiology technician – Miguel de Cervantes University (Aug 2019 – Jul 2020)
- B. S., Biology - University of Málaga, Spain (2018).

LANGUAGES

- Spanish, native
- English, fluent
- French, basic competence

PUBLICATIONS

- *Aliivibrio valkyriae* sp. nov. and *Aliivibrio boletzkyi* sp. nov. two new symbiotic species found in the light organ of *Sepiola atlantica*, **PA Pérez-Ferrer**, D Chillón-Pino, J Visitación, MK Nishiguchi *International Journal of Systematic and Evolutionary Microbiology* (in review).
- You talking to me? Quorum sensing in symbiotic microbes and their response to environmental variation, **PA Pérez-Ferrer** & MK Nishiguchi, *Microorganisms*, (in review).
- Genetic Variation in the Atlantic Bobtail Squid-Vibrio Symbiosis From the Galician Rías, **PA Pérez-Ferrer** M Ashraf, M Rodrigues, J Troncoso, MK Nishiguchi *Molecular Ecology* 34 (1) (2025).
- Microbial Biogeography, **PA Pérez-Ferrer** *Academic Press* (2024).
- Cephalopod-omics: emerging fields and technologies in cephalopod biology, T Baden, J Briseño, G Coffing, S Cohen-Bodénès... **PA Pérez-Ferrer** *Integrative and Comparative Biology* 63 (6), 1226-1239 (2023).
- Straightforward N-Acyl Homoserine Lactone Discovery and Annotation by LC–MS/MS-based Molecular Networking Alice MS Rodrigues, Raphaël Lami, Karine Escoubeyrou, Laurent Intertaglia, Clément Mazurek, Margot Doberva, **Pedro Pérez-Ferrer**, Didier Stien *Journal of proteome research* (2022).
- Quorum quenching strains isolated from the microbiota of sea anemones and holothurians attenuate *Vibrio coralliilyticus* virulence factors and reduce mortality in *Artemia salina* JC Reina, **P Pérez**, I Llamas *Microorganisms* 10 (3), 631 (2022).

CONFERENCES AND SEMINARS

- **2025 Molecular Investigation of Microbe-Host Interaction XXXVII** | Madison, WI June 2-4
Poster: Investigating Temperature-Dependent Functional Variation in the LuxR Protein from Symbiotic *Vibrio* Associated with Sepioids. *Authors: Pedro Antonio Pérez Ferrer, Michael Thompson, Michele K. Nishiguchi*
- **National Artificial Intelligence Research Resource Pilot, AI Unlocked: Empowering Higher Ed through Research and Discovery** | Denver, CO Apr 2-3. Workshop.
- **2024 Inclusive Excellence Symposium** | Merced, CA Nov 2024
Poster: Crossing communication barriers: Deciphering the molecular basis for autoinducer regulation between two symbiotic *Vibrio* species from *Sepiola* sp. squids. *Authors: P.A. Perez-Ferrer, R. Lami, M. Thompson and M.K Nishiguchi*
- **2024 DOE Joint Genome Institute (JGI) Annual Genomics of Energy & Environment Meeting** | Walnut Creek, CA Sept 30 – Oct 4
Poster: Improving the JGI fungal mitochondrial annotation pipeline. *Authors: Pedro Antonio Perez Ferrer, Sajeet Haridas, Asaf Salamov, Igor Grigoriev*
- **2024 Quantitative and Systems Biology Graduate Program Retreat** | Merced, CA Oct 29

Best Communication Award, **Poster:** Crossing communication barriers: Deciphering the molecular basis for autoinducer regulation between two symbiotic *Vibrio* species from *Sepiola* sp. squids. *Authors: P.A. Perez-Ferrer, R. Lami, M. Thompson and M.K Nishiguchi*

- **2024 CalTeach Summer Research Institute 2024** | Berkeley, CA Aug 8
Poster: Improving the JGI fungal mitochondrial annotation pipeline. *Authors: Pedro Antonio Perez Ferrer, Sajeet Haridas, Asaf Salamov, Igor Grigoriev*
- **2023 Quantitative and Systems Biology Graduate Program Retreat** | Merced, CA Oct 29
Poster: Shedding light on the *Sepiola-Vibrio* Symbiosis: a biochemical and genomic approach. *Authors: P.A. Perez-Ferrer, R. Lami and M.K Nishiguchi*
- **2023 Squid-Vibrio Meeting** | Merced, CA Jul 2-3
Talk: Crossing communication barriers: Autoinducer production between symbiotic *Vibrio* species from *Sepiola atlantica* and *Sepiola affinis* (Cephalopoda: Sepiolidae). *Authors: Pedro Antonio Pérez Ferrer, Raphaël Lami, Michele K. Nishiguchi*
- **2023 Yosemite Symbiosis Workshop** | Sierra Nevada Research Institute, Wawona, CA May 5-7
Talk: Crossing communication barriers: Autoinducer production between symbiotic *Vibrio* species from *Sepiola atlantica* and *Sepiola affinis* (Cephalopoda: Sepiolidae). *Authors: Pedro Antonio Pérez Ferrer, Raphaël Lami, Michele K. Nishiguchi*
- **2023 Society for Integrative and Comparative Biology Meeting (SICB)** | Austin, TX Jan 2-6
Talk: Crossing communication barriers: Autoinducer production between symbiotic *Vibrio* species from *Sepiola atlantica* and *Sepiola affinis* (Cephalopoda: Sepiolidae). *Authors: Pedro Antonio Pérez Ferrer, Raphaël Lami, Michele K. Nishiguchi*
- **2022 Yosemite Symbiosis Workshop** | Sierra Nevada Research Institute, Wawona, CA May 13-15
Poster: Genetic variation in the Atlantic bobtail squid - *Vibrio* symbiosis from the Galician Rías. *Authors: Pedro Antonio Pérez Ferrer, Maryam Ashraf, Marcelo Rodrigues, Jesús Troncoso and Michele K. Nishiguchi*
- **2022 Cephalopod International Advisory Council, Cephalopods in the Anthropocene** | Sesimbra, Portugal Apr 4-8
Poster: Genetic variation in the Atlantic bobtail squid - *Vibrio* symbiosis from the Galician Rías. *Authors: Pedro Antonio Pérez Ferrer, Maryam Ashraf, Marcelo Rodrigues, Jesús Troncoso and Michele K. Nishiguchi*
- **2019 National Meeting of Spanish Microbiology Society** | Málaga, Spain Jul 1-6
Poster: The microbiota of marine invertebrates as a source of quorum-extinguishing enzymes. *Authors: Pedro Antonio Pérez Ferrer, José Carlos Reina and Inmaculada Llamas*

GRANTS AND AWARDS

- Dr. Earl H. Myers & Ethel M. Myers, Oceanographic & Marine Biology Trust Fellowship.
- I Research and Transference Fellowship, University of Málaga.
- ERASMUS + European Programme Grant.
- NIH U-RISE Data Mentor Grant.
- Best Poster Presentation at the Quantitative and System Biology Retreat 2024.

COURSES AND ACCREDITATIONS

- **IBM AI Developer** | IBM, Sept, 2025. Credential ID: J0JSZQPR7W7F
In this Professional Certificate, learners acquired the core knowledge and practical skills to build web applications and generative AI powered solutions. They acquired a foundational understanding of the concepts related to programming and the Software Development Lifecycle (SDLC). They learned about the concepts, techniques, and applications related to AI, including generative AI and prompt engineering. The certificate holder has demonstrated their ability to design and develop web applications using HTML, CSS, JavaScript, Python, and Flask and build

chatbots and innovative solutions using generative AI-powered models, tools and technologies. The certificate holder is prepared for a promising career in AI development and software engineering, specializing in generative AI.

- **ACCESS-NAIRR AI Workshops** | University of Colorado Boulder, Apr 2-3, 2025. Verification link: <https://www.credly.com/badges/692036a3-b5ae-4640-b659-a72fdaa23e05>
This badge is awarded to individuals who have completed an ACCESS/NAIRR Pilot AI workshop, demonstrating engagement with national AI research infrastructure and computational resources.
Skills: Artificial Intelligence (AI), Cloud Computing for AI, Computational Research Methods, Cyberinfrastructure Literacy, Deep Learning, High Performance Computing (HPC), Information Technology, Leveraging AI
- **Bioinformatic Methods I** | University of Toronto, Sept 2020. Credential ID: QXEZZBFCFUXP
Skills: NCBI/Blast, Comparative Genomics, Multiple Alignment, Phylogeny, Metagenomics, RNAseq
- **Biology Meets Programming: Bioinformatics for Beginners** | UC San Diego, Jun 2020. Credential ID: VXFKBXQ5UT6C
Skills: Linux Command Line, Basic Python, Basic R, Genomic Comparison tools.

VOLUNTEERING

- **Outreach Activities:** multiple participation in tables showcasing the research about Dr. Nishiguchi's lab to general audiences at Bobcat Day at UC Merced and Science Day in Stanislaus University.
- *Skills: general audience science communication, community engagement, public outreach, simplification of complex scientific concepts.*
- **Recovery** of coastal ecosystems in the Maro-Cerro Gordo Cliffs Nature Reserve.
- **Recovery** of Quejigal and Pinsapar from the Sierra de las Nieves. Construction of shelters for amphibian fauna.
- **Participation** in the coastal cleanup campaigns organized in Playas del Palo and Baños del Carmen
- *Skills: Ecological evaluation of the situation of the local fauna and flora, Study of the presence of invasive or exotic species.*

PROFESSIONAL AFFILIATIONS

- Belongs to Sigma Xi, The Scientific Research Honor Society (Member).
- Spanish Society of Microbiology (SEM) (Member).
- Society of Integrative and Comparative Biology (SICB) (Member).